

Лабораторная работа №6

Адресация IPv4 и IPv6. Двойной стек

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Цель работы

Изучение принципов распределения и настройки адресного пространства на устройствах сети, а также реализация сети с двойным стеком протоколов (IPv4 и IPv6).

Задачи

1. Расчет подсетей IPv4 (VLSM) и IPv6.
2. Реализация топологии сети в GNS3.
3. Настройка маршрутизатора FRR (IPv4).
4. Настройка маршрутизатора VyOS (IPv6).
5. Конфигурирование конечных устройств.
6. Анализ трафика и проверка связности.

Расчетная часть

Задача: Разбить сеть `172.16.20.0/24` на 3 подсети (126, 62, 62 узла).

Решение:

1. **126 узлов:** Префикс `/25` (128 адресов).

- Сеть: `172.16.20.0/25`

2. **62 узла:** Префикс `/26` (64 адреса).

- Сеть: `172.16.20.128/26`

3. **62 узла:** Префикс `/26` (64 адреса).

- Сеть: `172.16.20.192/26`

Задача: Разбить сеть `2001:db8:c0de::/48`.

Решение:

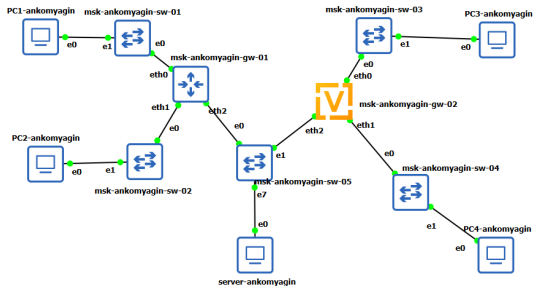
Используем идентификатор подсети (Subnet ID):

1. Подсеть 1: `2001:db8:c0de:1::/64`
2. Подсеть 2: `2001:db8:c0de:2::/64`
3. ...
4. Подсеть N: `2001:db8:c0de:ffff::/64`

Реализация топологии

Сеть состоит из двух сегментов:

- Левый сегмент: IPv4 (Router FRR)
- Правый сегмент: IPv6 (Router VyOS)
- Центральный сервер: Dual Stack



Настройка IPv4

Настройка IP-адресов и шлюзов на PC1 и PC2.

```
PC1-ankomyagin - PuTTY
Hostname is too long. (Maximum 12 characters)

VPCS> ip 172.16.20.10/25 172.16.20.1
Checking for duplicate address...
VPCS : 172.16.20.10 255.255.255.128 gateway 172.16.20.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 172.16.20.10/25
GATEWAY     : 172.16.20.1
DNS         :
MAC         : 00:50:79:66:68:02
LPORT      : 10026
RHOST:PORT  : 127.0.0.1:10027
MTU         : 1500

VPCS> show ipv6

NAME       : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6802/64
GLOBAL SCOPE    :
DNS             :
ROUTER LINK-LAYER :
MAC             : 00:50:79:66:68:02
LPORT          : 10026
```

Настройка маршрутизатора FRR

Настройка интерфейсов eth0, eth1, eth2 на маршрутизаторе msk-ankomyagin-gw-01.

```
msk-ankomyagin-gw-01 - PuTTY
Started watchfrr
 * Starting sshd ... [ ok ]

Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname msk-ankomyagin-gw-01
msk-ankomyagin-gw-01(config)# exit
msk-ankomyagin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-01# configure terminal
msk-ankomyagin-gw-01(config)# interface eth0
msk-ankomyagin-gw-01(config-if)# ip address 172.16.20.1/25
msk-ankomyagin-gw-01(config-if)# no shutdown
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# interface eth1
msk-ankomyagin-gw-01(config-if)# ip address 172.16.20.129/25
msk-ankomyagin-gw-01(config-if)# no shutdown
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# interface eth2
msk-ankomyagin-gw-01(config-if)# ip address 64.100.1.1/24
msk-ankomyagin-gw-01(config-if)# no shutdown
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# show running-config
% Unknown command: show running-config
msk-ankomyagin-gw-01(config)# write memory
% Unknown command: write memory
msk-ankomyagin-gw-01(config)# exit
```

Проверка связности IPv4

Успешное прохождение эхо-запросов (Ping) и трассировки маршрута.

```
PC1-ankomyagin - PuTTY
GATEWAY      : 172.16.20.1
DNS           :
MAC          : 00:50:79:66:68:02
LPORT        : 10024
RHOST:PORT    : 127.0.0.1:10025
MTU          : 1500

VPCS> ping 172.16.20.138/25

84 bytes from 172.16.20.138 icmp_seq=1 ttl=63 time=16.867 ms
84 bytes from 172.16.20.138 icmp_seq=2 ttl=63 time=17.073 ms
84 bytes from 172.16.20.138 icmp_seq=3 ttl=63 time=9.869 ms
84 bytes from 172.16.20.138 icmp_seq=4 ttl=63 time=8.658 ms
84 bytes from 172.16.20.138 icmp_seq=5 ttl=63 time=14.852 ms

VPCS> ping 64.100.1.10/24

84 bytes from 64.100.1.10 icmp_seq=1 ttl=63 time=19.613 ms
84 bytes from 64.100.1.10 icmp_seq=2 ttl=63 time=6.597 ms
84 bytes from 64.100.1.10 icmp_seq=3 ttl=63 time=3.336 ms
84 bytes from 64.100.1.10 icmp_seq=4 ttl=63 time=15.408 ms
84 bytes from 64.100.1.10 icmp_seq=5 ttl=63 time=1.797 ms

VPCS> █

PC2-ankomyagin - PuTTY
GATEWAY      : 172.16.20.129
DNS           :
MAC          : 00:50:79:66:68:01
LPORT        : 10022
RHOST:PORT    : 127.0.0.1:10023
MTU          : 1500

VPCS> ping 172.16.20.10/25

84 bytes from 172.16.20.10 icmp_seq=1 ttl=63 time=7.327 ms
84 bytes from 172.16.20.10 icmp_seq=2 ttl=63 time=9.642 ms
84 bytes from 172.16.20.10 icmp_seq=3 ttl=63 time=3.010 ms
84 bytes from 172.16.20.10 icmp_seq=4 ttl=63 time=4.503 ms
84 bytes from 172.16.20.10 icmp_seq=5 ttl=63 time=3.431 ms

VPCS> ping 64.100.1.10/24

84 bytes from 64.100.1.10 icmp_seq=1 ttl=63 time=4.648 ms
84 bytes from 64.100.1.10 icmp_seq=2 ttl=63 time=2.338 ms
84 bytes from 64.100.1.10 icmp_seq=3 ttl=63 time=8.214 ms
84 bytes from 64.100.1.10 icmp_seq=4 ttl=63 time=5.688 ms
84 bytes from 64.100.1.10 icmp_seq=5 ttl=63 time=4.722 ms

VPCS> █
```

Настройка IPv6

Конфигурация маршрутизатора VyOS

Настройка интерфейсов и префиксов сети на маршрутизаторе msk-ankomyagin-gw-02.

```
msk-ankomyagin-gw-02 login: vyos
Password:
Welcome to VyOS!

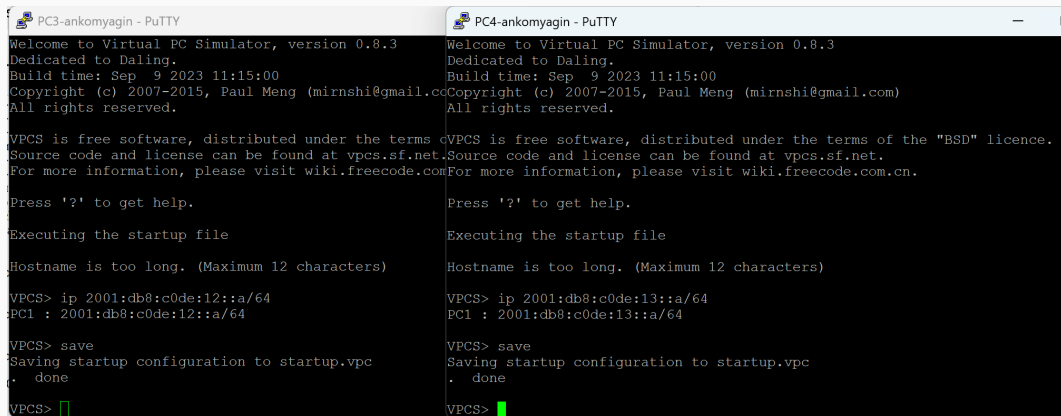
Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@msk-ankomyagin-gw-02:~$ configure
[edit]
2::1/64k-ankomyagin-gw-02# set interfaces ethernet eth 0 address 2001:db8:c0de:1
2001:db8:c0de:1::1/64
Configuration path: interfaces ethernet eth [0] is not valid
Set failed
2001:db8:c0de:1::1/64
[edit]
vyos@msk-ankomyagin-gw-02# set interfaces ethernet eth0 address 2001:db8:c0de:12::1/64
[edit]
3::1/64k-ankomyagin-gw-02# set interfaces ethernet eth1 address 2001:db8:c0de:1
2001:db8:c0de:1::1/64
[edit]
vyos@msk-ankomyagin-gw-02#
[edit]
2001:db8:c0de:1::1/64k-ankomyagin-gw-02# set service router-advert interface eth1 prefix 2001:
2001:db8:c0de:1::1/64
[edit]
2001:db8:c0de:1::1/64sk-ankomyagin-gw-02# set interfaces ethernet eth2 address 2001:db8:c0de:11
2001:db8:c0de:11::1/64
[edit]
2001:db8:c0de:11::1/64gin-gw-02# set service router-advert interface eth2 prefix 2001:
2001:db8:c0de:11::1/64
[edit]
vyos@msk-ankomyagin-gw-02#
```

Конфигурация конечных устройств IPv6

Настройка статических IPv6 адресов на PC3 и PC4.



```
PC3-ankomyagin - PuTTY
Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 2001:db8:c0de:12::a/64
PC1 : 2001:db8:c0de:12::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> █

PC4-ankomyagin - PuTTY
Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 2001:db8:c0de:13::a/64
PC1 : 2001:db8:c0de:13::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> █
```

Проверка связности IPv6

Тестирование доступности узлов и маршрутизации в сегменте IPv6.

```
PC3-ankomyagin - PuTTY
Executing the startup file

PC1 : 2001:db8:c0de:12::a/64

VPCS> ping 2001:db8:c0de:13::a/64

2001:db8:c0de:13::a icmp6_seq=1 ttl=62 time=28.655 ms
2001:db8:c0de:13::a icmp6_seq=2 ttl=62 time=4.184 ms
2001:db8:c0de:13::a icmp6_seq=3 ttl=62 time=4.178 ms
2001:db8:c0de:13::a icmp6_seq=4 ttl=62 time=4.179 ms
2001:db8:c0de:13::a icmp6_seq=5 ttl=62 time=5.027 ms

VPCS> trace 2001:db8:c0de:13::a/64

trace to 2001:db8:c0de:13::a, 64 hops max
 1 2001:db8:c0de:12::1 9.283 ms 4.526 ms 1.584 ms
 2 2001:db8:c0de:13::a 6.144 ms 2.056 ms 3.975 ms

VPCS> ping 2001:db8:c0de:11::a/64

2001:db8:c0de:11::a icmp6_seq=1 ttl=62 time=4.159 ms
2001:db8:c0de:11::a icmp6_seq=2 ttl=62 time=3.330 ms
2001:db8:c0de:11::a icmp6_seq=3 ttl=62 time=4.750 ms
2001:db8:c0de:11::a icmp6_seq=4 ttl=62 time=2.775 ms
2001:db8:c0de:11::a icmp6_seq=5 ttl=62 time=5.383 ms

VPCS> trace 2001:db8:c0de:11::a/64

trace to 2001:db8:c0de:11::a, 64 hops max
 1 2001:db8:c0de:12::1 1.968 ms 2.947 ms 2.255 ms
 2 2001:db8:c0de:11::a 3.956 ms 4.704 ms 6.778 ms

VPCS> █

PC4-ankomyagin - PuTTY
Executing the startup file

PC1 : 2001:db8:c0de:13::a/64

VPCS> ping 2001:db8:c0de:12::a/64

2001:db8:c0de:12::a icmp6_seq=1 ttl=62 time=5.074 ms
2001:db8:c0de:12::a icmp6_seq=2 ttl=62 time=5.476 ms
2001:db8:c0de:12::a icmp6_seq=3 ttl=62 time=4.235 ms
2001:db8:c0de:12::a icmp6_seq=4 ttl=62 time=2.819 ms
2001:db8:c0de:12::a icmp6_seq=5 ttl=62 time=5.706 ms
2001:db8:c0de:12::a icmp6_seq=6 timeout

VPCS> trace 2001:db8:c0de:12::a/64

trace to 2001:db8:c0de:12::a, 64 hops max
 1 2001:db8:c0de:13::1 5.031 ms 1.330 ms 1.280 ms
 2 2001:db8:c0de:12::a 6.789 ms 4.574 ms 2.962 ms

VPCS> ping 2001:db8:c0de:11::a/64

2001:db8:c0de:11::a icmp6_seq=1 ttl=62 time=6.217 ms
2001:db8:c0de:11::a icmp6_seq=2 ttl=62 time=3.334 ms
2001:db8:c0de:11::a icmp6_seq=3 ttl=62 time=5.011 ms
2001:db8:c0de:11::a icmp6_seq=4 ttl=62 time=3.615 ms
2001:db8:c0de:11::a icmp6_seq=5 ttl=62 time=3.433 ms

VPCS> trace 2001:db8:c0de:11::a/64

trace to 2001:db8:c0de:11::a, 64 hops max
 1 2001:db8:c0de:13::1 2.059 ms 1.271 ms 1.290 ms
 2 2001:db8:c0de:11::a 4.725 ms 4.555 ms 4.517 ms

VPCS> █
```

Анализ трафика

Анализ дампа трафика на линке сервера:

- Видны запросы **ARP** для разрешения IPv4 адресов.
- Видны запросы **Neighbor Solicitation (ICMPv6)** для IPv6.

Захват с Standard input [server-anonymycin Ethernet0 to msk-anonymycin-sw-05 Ethernet7]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

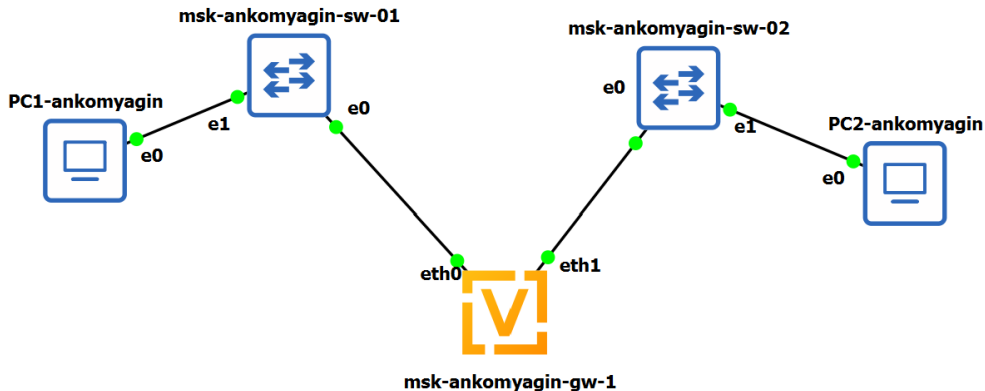
Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
5	4.003031	2001:db8:c0de:11::a	2001:db8:c0de:12::	ICMPv6	118	Echo (ping) request id=0xe9a1, seq=3, hop limit=64 (no response f
6	4.006061	2001:db8:c0de:11::1	2001:db8:c0de:11::a	ICMPv6	118	Echo (ping) reply id=0xe9a1, seq=3, hop limit=64
7	5.398219	fe80::ee3:e1ff:fe4...	2001:db8:c0de:11::a	ICMPv6	86	Neighbor Solicitation for 2001:db8:c0de:11::a from 0c:e3:e1:f4:06
8	6.004708	2001:db8:c0de:11::a	2001:db8:c0de:12::	ICMPv6	118	Echo (ping) request id=0xe9a1, seq=4, hop limit=64 (no response f
9	6.007893	2001:db8:c0de:11::1	2001:db8:c0de:11::a	ICMPv6	118	Echo (ping) reply id=0xe9a1, seq=4, hop limit=64
10	6.412964	fe80::ee3:e1ff:fe4...	2001:db8:c0de:11::a	ICMPv6	86	Neighbor Solicitation for 2001:db8:c0de:11::a from 0c:e3:e1:f4:06
11	7.438883	fe80::ee3:e1ff:fe4...	2001:db8:c0de:11::a	ICMPv6	86	Neighbor Solicitation for 2001:db8:c0de:11::a from 0c:e3:e1:f4:06
12	8.007741	2001:db8:c0de:11::a	2001:db8:c0de:12::	ICMPv6	118	Echo (ping) request id=0xe9a1, seq=5, hop limit=64 (no response f
13	8.009386	2001:db8:c0de:11::1	2001:db8:c0de:11::a	ICMPv6	118	Echo (ping) reply id=0xe9a1, seq=5, hop limit=64
14	69.915656	fe80::ee3:e1ff:fe4...	ff02::1:ff00:a	ICMPv6	86	Neighbor Solicitation for 2001:db8:c0de:11::a from 0c:e3:e1:f4:06
15	69.916252	2001:db8:c0de:11::a	fe80::ee3:e1ff:fe4...	ICMPv6	86	Neighbor Advertisement 2001:db8:c0de:11::a (sol, ovr) is at 00:56
16	69.918628	2001:db8:c0de:12::a	2001:db8:c0de:11::a	ICMPv6	118	Echo (ping) request id=0x2fa2, seq=1, hop limit=63 (reply in 17)
17	69.920222	2001:db8:c0de:11::a	2001:db8:c0de:12::a	ICMPv6	118	Echo (ping) reply id=0x2fa2, seq=1, hop limit=63 (request in 16)
18	70.928546	2001:db8:c0de:12::a	2001:db8:c0de:11::a	ICMPv6	118	Echo (ping) request id=0x2fa2, seq=2, hop limit=63 (reply in 19)
19	70.929859	2001:db8:c0de:11::a	2001:db8:c0de:12::a	ICMPv6	118	Echo (ping) reply id=0x2fa2, seq=2, hop limit=63 (request in 18)
20	71.934631	2001:db8:c0de:12::a	2001:db8:c0de:11::a	ICMPv6	118	Echo (ping) request id=0x2fa2, seq=3, hop limit=63 (reply in 21)
21	71.935120	2001:db8:c0de:11::a	2001:db8:c0de:12::a	ICMPv6	118	Echo (ping) reply id=0x2fa2, seq=3, hop limit=63 (request in 20)
22	72.939462	2001:db8:c0de:12::a	2001:db8:c0de:11::a	ICMPv6	118	Echo (ping) request id=0x2fa2, seq=4, hop limit=63 (reply in 23)
23	72.939712	2001:db8:c0de:11::a	2001:db8:c0de:12::a	ICMPv6	118	Echo (ping) reply id=0x2fa2, seq=4, hop limit=63 (request in 22)
24	73.946776	2001:db8:c0de:12::a	2001:db8:c0de:11::a	ICMPv6	118	Echo (ping) request id=0x2fa2, seq=5, hop limit=63 (reply in 25)
25	73.947139	2001:db8:c0de:11::a	2001:db8:c0de:12::a	ICMPv6	118	Echo (ping) reply id=0x2fa2, seq=5, hop limit=63 (request in 24)
26	97.833814	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) request id=0x4ba2, seq=1/256, ttl=63 (reply in 29)
27	97.834176	Private_66:68:00	Broadcast	ARP	64	Who has 64.100.1.1? Tell 64.100.1.10
28	97.834866	0c:c6:a1:41:00:02	Private_66:68:00	ARP	60	64.100.1.1 is at 0c:c6:a1:41:00:02
29	97.837916	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) reply id=0x4ba2, seq=1/256, ttl=64 (request in 26)
30	98.844604	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) request id=0x4ca2, seq=2/512, ttl=63 (reply in 31)
31	98.845819	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) reply id=0x4ca2, seq=2/512, ttl=64 (request in 30)
32	99.853197	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) request id=0x4da2, seq=3/768, ttl=63 (reply in 33)
33	99.854088	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) reply id=0x4da2, seq=3/768, ttl=64 (request in 32)

Самостоятельная работа

Топология и задача

Реализация топологии “галочка” с одним маршрутизатором VyOS, разделяющим сеть на две подсети с поддержкой Dual Stack.



Конфигурация интерфейсов маршрутизатора VyOS для работы с двумя подсетями одновременно по IPv4 и IPv6.

![]image/(19.png)

Проверка результатов

Успешная проверка связности между подсетями по обоим протоколам.

```
PC1-ankomyagin - PuTTY
Checking for duplicate address...
VPCS : 10.10.1.98 255.255.255.224 gateway 10.10.1.97

PC1 : 2001:db8:1:1::a/64

VPCS> clear ip
IPv4 address/mask, gateway, DNS, and DHCP cleared

VPCS> clear ipv6
IPv6 address/mask and router link-layer address cleared

VPCS> ip 10.10.1.98/27 10.10.1.97
Checking for duplicate address...
VPCS : 10.10.1.98 255.255.255.224 gateway 10.10.1.97

VPCS> ip 2001:db8:1:1::a/64
PC1 : 2001:db8:1:1::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 10.10.1.98/27
GATEWAY    : 10.10.1.97
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10002
RHOST:PORT : 127.0.0.1:10003
MTU        : 1500

VPCS> show ipv6

PC2-ankomyagin - PuTTY
DNS      :
ROUTER LINK-LAYER :
MAC      : 00:50:79:66:68:01
LPORT    : 10000
RHOST:PORT : 127.0.0.1:10001
MTU      : 1500

VPCS> ping 10.10.1.98
84 bytes from 10.10.1.98 icmp_seq=1 ttl=63 time=10.375 ms
84 bytes from 10.10.1.98 icmp_seq=2 ttl=63 time=5.439 ms
84 bytes from 10.10.1.98 icmp_seq=3 ttl=63 time=6.371 ms
84 bytes from 10.10.1.98 icmp_seq=4 ttl=63 time=7.205 ms
84 bytes from 10.10.1.98 icmp_seq=5 ttl=63 time=5.816 ms

VPCS> trace 10.10.1.98
trace to 10.10.1.98, 8 hops max, press Ctrl+C to stop
 1 10.10.1.17 4.084 ms 2.969 ms 2.159 ms
 2 *10.10.1.98 4.789 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> ping 2001:db8:1:1::a/64
2001:db8:1:1::a icmp6_seq=1 ttl=62 time=3.379 ms
2001:db8:1:1::a icmp6_seq=2 ttl=62 time=7.749 ms
2001:db8:1:1::a icmp6_seq=3 ttl=62 time=4.993 ms
2001:db8:1:1::a icmp6_seq=4 ttl=62 time=6.354 ms
2001:db8:1:1::a icmp6_seq=5 ttl=62 time=5.893 ms

VPCS> trace 2001:db8:1:1::a/64
trace to 2001:db8:1:1::a, 64 hops max
 1 2001:db8:1:4::1 8.425 ms 1.881 ms 2.984 ms
 2 2001:db8:1:1::a 8.770 ms 5.028 ms 5.354 ms

VPCS>
```

Выводы

- Изучены методы бесклассовой адресации (CIDR) и VLSM.
- Приобретены навыки настройки оборудования FRR и VyOS.
- Реализована и протестирована сеть с двойным стеком протоколов.
- Проведен анализ различий в служебных протоколах IPv4 и IPv6 (ARP vs NDP).